WalletConnect Network 2.0: A Federated Key-Value Store with Blockchain Consensus

Nodar Darksome[1] Ivan Reshetnikov[1] Rafael Quintero[1] Ivan Hoy West[1] Derek Rein[1]

[1] *Special thanks to Daniel Martinez*

Abstract

WalletConnect Network 2.0 (WCN 2.0) is a federated, low-latency key-value store operated by over 20 independent node operators. Building on WCN 1.0, the system introduces regionalized quorums to minimize latency for use cases that do not require global replication, while preserving global state when necessary. WCN 2.0 replaces Raft consensus with a blockchain-based consensus, making it the first distributed key-value store to leverage a blockchain ledger for coordination. The architecture further separates into WCN Database (storage) and WCN Node (coordination), enabling independent deployments and paving the way for elastic scaling. By decentralizing coordination across more regions and operators, WCN 2.0 strengthens resilience, improves performance, and advances WalletConnect's mission of building fully permissionless applications.

1 Introduction

The WalletConnect Network serves as critical infrastructure for decentralized applications, providing a real-time communication layer for wallet-to-dapp interactions. As the ecosystem has grown to support millions of users across diverse geographic regions, the need for a more scalable, resilient, and performant infrastructure has become apparent.

WCN 1.0 successfully demonstrated the viability of a federated key-value store using Raft consensus [8]. However, operational experience revealed several limitations: (1) global replication introduced unnecessary latency for region-specific workloads, (2) Raft's leader-based architecture created operational complexity at scale, and (3) the monolithic design made independent scaling of storage and coordination difficult.

WCN 2.0 addresses these challenges through three architectural innovations: regionalized quorums for latency optimization, blockchain-based consensus for verifiable coordination, and separation of database and coordinator nodes for independent scaling.

1.1 Motivation

The WalletConnect Protocol remains the primary application of the WalletConnect Network, alongside Smart Sessions. In practice, most protocol workloads do not require global replication of state; rather, they benefit from regionalized quorums that can serve client requests with lower round-trip latency while still preserving the option of global coordination when required.

Operational experience with WCN 1.0 also revealed challenges in managing Raft consensus at scale. While effective in a permissioned environment, Raft introduces operational complexity and does not naturally extend to a permissionless model. For workloads characterized by frequent reads and infrequent writes, a blockchain-based ledger provides a more natural coordination layer: it is tamper-evident, globally verifiable, and eliminates the need for a single trusted leader.

Finally, the architectural split between WCN Database (storage) and WCN Node (coordinator) addresses limitations encountered in earlier deployments. Independent upgrade cycles and rolling deployments reduce operational risk and pave the way for elastic scaling of the storage layer without disrupting consensus. Together, these design changes bring WCN 2.0 closer to its long-term goal: a high-performance, decentralized, and permissionless infrastructure for real-time applications.

1.2 Contributions

The main contributions of WCN 2.0 are:

- **Regionalized quorums** that reduce latency by up to 60% for geographically localized workloads while maintaining global consistency when needed
- **Blockchain-based consensus** replacing Raft, making WCN 2.0 the first distributed key-value store to use a blockchain ledger for coordination
- **Coordinator-Database separation** enabling independent deployments, rolling upgrades, and elastic scaling of storage

- **Enhanced decentralization** with over 20 independent operators across multiple regions, improving resilience and advancing toward permissionless operation

2 Background and Related Work

2.1 Distributed Key-Value Stores

Distributed key-value stores form the backbone of modern cloud infrastructure. Systems like Amazon DynamoDB [4], Apache Cassandra [6], and etcd [5] have demonstrated various approaches to achieving consistency, availability, and partition tolerance. Most production systems rely on consensus protocols like Paxos [7] or Raft [8] for coordination.

WCN 1.0 followed this established pattern, using Raft consensus across a federated set of operators. While Raft provides strong consistency guarantees and is well-understood operationally, it requires a stable leader and does not naturally accommodate permissionless participation—both limitations that motivated the redesign in WCN 2.0.

2.2 Blockchain as Coordination Layer

Blockchain systems have traditionally been associated with cryptocurrency applications, but recent work has explored their use as coordination mechanisms for distributed systems [1]. The append-only, cryptographically verifiable nature of blockchain ledgers provides natural advantages for audit trails and Byzantine fault tolerance.

To our knowledge, WCN 2.0 represents the first production deployment of a distributed key-value store that uses a blockchain ledger as its primary coordination mechanism. This approach trades some throughput for verifiability and eliminates single points of failure inherent in leader-based consensus.

2.3 Geo-Distributed Systems

Geo-distributed databases like Google Spanner [3] and CockroachDB [2] have explored various strategies for managing latency across regions. These systems typically use techniques like quorum-based replication, synchronized clocks, or explicit user control over data placement.

WCN 2.0's approach to regionalized quorums differs by making regional vs. global replication an implicit decision based on workload patterns rather than explicit user configuration. This design choice reflects the specific requirements of the WalletConnect Protocol, where most interactions are inherently regional but occasional global coordination is necessary.

3 Architecture

WCN 2.0 introduces three major architectural changes: regional clusters, heterogeneous nodes and Smart-Contract-based consensus. This section describes each of these changes in detail.

3.1 Regional Clusters

In WCN 1.0 all storage operations were replicated to global replica sets. A replica set consisted of 3 nodes, one node per geographic region. Both reads and writes required a majority quorum within the replica set to be considered successful. This approach provided strong consistency and fault tolerance, however operation execution latencies were sub-optimal for region-specific workloads. Addition of new regions to the network wasn't trivial, as that required cross-regional data rebalancing.

WCN 2.0 splits the network into regional clusters. All nodes within a regional cluster are topologically (and usually geographically) co-located. Replication factor has also been increased to 5, meaning replica sets now consist of 5 independent storage backends. Clusterization does improve operation execution latency, however it also reduces the fault tolerance of the network making it more susceptible to natural and infrastructural disasters. To overcome this the clusterization is being constrained in a way so the deployments are not located too close to each other. The factor of clusterization is the target network latency beetwen regional deployments being in the 30-50ms range.

3.2 Heterogeneous Nodes

WCN 1.0 nodes were homogeneous, performing replication coordination, data storage/retrieval and consensus management. With this design horizontal scaling was only possible via adding more stateful nodes into the network, which required data rebalancing. Node upgrades were also non-trivial, requiring a separate orchestration machinery to make sure that only limited amount of nodes were offline at any given time.

In WCN 2.0 the storage logic that interacts with the filesystem has been extracted into a separate executable. This change made majority of WCN nodes stateless, simplifying horizontal scaling and upgrades.

3.3 Smart-Contract-based Consensus

WCN 2.0 replaces Raft with a Smart-Contract-based consensus layer. For each regional cluster a separate Smart-Contract instance is being deployed on the Optimism chain. Nodes use the on-chain state as the source of truth regarding the current configuration of the cluster.

The Smart-Contact logic is responsible for management of the following:

- **Permissions**: Each organization operating WCN nodes must have an on-chain record within the scope of the regional cluster Smart-Contract. Creation and removal of these records effectively manage the permissions of which deployments are allowed to communicate with other deployments in the cluster.

- **Service discovery**: Node operators can update their on-chain state in order to specify the current list of the nodes they host, IP addresses of their deployments, public keys, etc.

- **Data rebalancing**: Each time a node operator deployment is being added or removed the data needs to be rebalanced within the cluster. The progress of this rebalancing is being tracked within the Smart-Contract state.

- **Maintenance**: When a node operator needs to perform a disruptive infrastructural change, which makes their deployment temporary unavailable, they must trigger a Smart-Contract call to notify the cluster about it. The Smart-Contract here acts as a semaphore limiting how many deployments can be under maintenance at any given time.

4 Performance and Benefits

4.1 Latency Improvements

We evaluated WCN 2.0's latency characteristics using production traffic patterns over a 30-day period. Table 1 compares write latencies between WCN 1.0 (global replication) and WCN 2.0 (regionalized quorums).

Table 1: Write Latency Comparison (p50/p95/p99 in ms)

Region	WCN 1.0	WCN 2.0	Improvement
North America	145/280/420	52/95/140	64% (p50)
Europe	165/310/485	58/105/165	65% (p50)
Asia-Pacific	180/340/520	65/115/180	64% (p50)
Cross-region	220/410/650	215/405/640	2% (p50)

Regional writes show consistent improvement of 60-65% in median latency across all regions. Cross-region writes (requiring global quorum) show minimal degradation, validating that the promotion mechanism correctly identifies and handles global state.

Read latency remains unchanged from WCN 1.0, as both systems serve reads from local database replicas. The 99th percentile read latency averages 15ms across all regions.

4.2 Operational Benefits

The coordinator-database separation has significantly improved operational agility:

- **Rolling upgrades**: Database upgrades can be performed with zero downtime by rotating replicas out of service. Coordinator upgrades benefit from the blockchain's fault tolerance, allowing gradual rollout across operators.

- **Deployment frequency**: Average deployment frequency increased from 2-3 per month in WCN 1.0 to 8-12 per month in WCN 2.0, reflecting reduced deployment risk.

- **Incident recovery**: Mean time to recovery (MTTR) for database failures decreased from 8 minutes to 2 minutes due to faster failover to healthy replicas.

4.3 Security and Resilience

WCN 2.0's decentralized architecture provides several security advantages:

Byzantine fault tolerance: The blockchain consensus tolerates up to $\frac{n-1}{3}$ Byzantine failures among coordinator nodes. With 20+ operators, the system can withstand 6+ malicious or faulty coordinators.

Audit trail: All state transitions are recorded in the blockchain with cryptographic proofs. This provides complete traceability for debugging and security analysis.

Geographic resilience: With 8+ regions, the system remains operational during complete regional failures. Testing confirmed availability during simulated loss of any two regions simultaneously.

No single point of failure: Unlike WCN 1.0's Raft leader, WCN 2.0 has no privileged nodes. Coordinator failures are handled transparently by the blockchain consensus.

5 Hardfork Migration Strategy

In WCN, every record is subject to a maximum time-to-live (TTL) of 30 days. This constraint effectively transforms migration from a stateful data transfer problem into a controlled process of state convergence, where only live records are relevant. After one complete cycle of shadowing, the new storage will have fully converged with the legacy system, enabling a seamless transition without manual data migration.

Phase 1 - WCN 2.0 Initial Deployment

The migration begins with a controlled rollout of WCN 2.0 under WalletConnect's direct operation

- All nodes operated by WalletConnect: To ensure stability and consistency during the initial deployment, all participating nodes are managed centrally. This greatly simplifies monitoring, profiling and deploying updates to the cluster as needed.
- EU cluster acts as global: A single cluster deployed to one of the Central European datacenters is designated as the global cluster, serving as the authoritative source of truth for all records.
- Begin shadowing writes: Incoming writes are duplicated into the WCN 2.0 cluster while continuing to serve consumers from WCN 1.0. This allows validation of write-path correctness without impacting production traffic and begins the 30-day period of state convergence.
- Performance Profiling: Latency, throughput, and consistency metrics are collected to evaluate the behavior of WCN 2.0 under production-like workloads.

Phase 2 - Inviting Third Parties into the Network

Once the system demonstrates stability under WalletConnect's operation, external participants are gradually invited.

- Controlled Onboarding: Third-party operators are integrated into the cluster under strict governance to ensure adherence to protocol specifications.
- Data Migrations within the Cluster: As new nodes join, the data stored in the system is redistributed among the nodes to rebalance the keyspace and ensure uniform load across the cluster.

Phase 3 - Shadowing Reads

After increasing the network capacity by adding enough third party operators, the migration progresses to shadowing reads:

- Gradual Enablement: The amount of consumer read requests is gradually increased, while monitoring the network's performance.
- Continuous Profiling: Read-path performance is analyzed to detect potential bottlenecks or divergence in state.

Phase 4 - Switching Consumers to WCN 2.0 Global Cluster

When monitoring confirms that there are no outstanding issues remain, and no state divergence exists between WCN 1.0 and WCN 2.0, consumers are switched to use the WCN 2.0 global cluster as their primary storage. This marks the first major milestone in the migration process, where WCN 2.0 assumes production responsibility. At this point, both WCN 2.0 and WCN 1.0 clusters are fully operational. Consumers keep writing data to both clusters, allowing rollback if critical issues emerge.

Phase 5 - Deployment of Regional Clusters

Regional clusters are deployed, and third party operators are invited to join them:

- Request Shadowing: Writes and reads to the global cluster are also shadowed into regional clusters to validate their performance under real-world traffic.
- Capacity Verification: Regional clusters are stress-tested to ensure they can handle peak workloads without degradation.

Phase 6 - Enabling Regionalized Storage

Once regional clusters demonstrate stability, consumers are updated to enable full use of regional clusters to improve storage latency and user experience.

Phase 7 - Decommissioning WCN 1.0

The final stage of migration involves retiring the legacy system. Once all consumers have successfully switched to WCN 2.0, WCN 1.0 is fully decommissioned.

6 Updated Reward System

WCN 2.0 introduces an updated reward mechanism for node operators that better aligns incentives with system health and performance. The new system incorporates several improvements over WCN 1.0:

Performance-based rewards: Operator rewards are adjusted based on node uptime, latency performance, and consensus participation. Operators with consistent high performance receive proportionally higher rewards.

Token-price-adjusted rewards: To maintain stable incentives in the face of token price volatility, rewards are dynamically adjusted based on a moving average of token price over the previous 30 days. This ensures operators receive consistent economic value regardless of short-term price fluctuations.

Geographic diversity incentives: Additional rewards are allocated to operators in underserved regions to encourage global distribution. This strengthens network resilience and reduces latency for users worldwide.

The specific reward formula and distribution parameters are subject to ongoing governance and may be adjusted based on operational experience and community feedback.

7 Future Work

Several directions for future development have been identified:

7.1 Autoscaling Storage Layer

The current coordinator-database separation enables elastic scaling of storage, but autoscaling is not yet implemented. Future work will develop policies for automatic database replica scaling based on load metrics, further improving operational efficiency.

7.2 Permissionless Operation

While WCN 2.0's blockchain consensus is compatible with permissionless operation, the current deployment remains permissioned with vetted operators. The path to permissionless operation requires:

- Stake-based sybil resistance mechanisms
- Reputation systems to identify and penalize malicious actors
- Dynamic operator sets that can adapt as nodes join and leave
- Enhanced economic security models

7.3 Ecosystem Use Cases

Beyond WalletConnect Protocol and Smart Sessions, WCN 2.0's architecture may serve other decentralized applications requiring low-latency, verifiable state coordination. Potential use cases include:

- Decentralized identity systems
- Cross-chain messaging protocols
- Real-time collaborative applications
- IoT device coordination

7.4 Governance Updates

The reward system and operational parameters will evolve based on community governance. Future governance mechanisms may include:

- On-chain voting for parameter adjustments
- Operator councils for technical decision-making
- Transparent metrics dashboards for community oversight

8 Conclusion

WalletConnect Network 2.0 represents a significant evolution in distributed key-value store design, introducing three major innovations: regionalized quorums for latency optimization, blockchain-based consensus for verifiable coordination, and coordinator-database separation for operational agility.

Production deployment demonstrates 60-65% latency reduction for regional workloads while maintaining strong consistency guarantees. The expanded operator set across 20+ independent entities and 8+ geographic regions significantly improves resilience and advances WalletConnect's decentralization goals.

By replacing traditional consensus mechanisms with blockchain-based coordination, WCN 2.0 establishes a new architectural pattern for distributed systems that require both high performance and verifiable state management. The system serves as foundational infrastructure for WalletConnect's mission to build fully permissionless, globally accessible decentralized applications.

As the first production distributed key-value store to leverage blockchain for consensus, WCN 2.0 demonstrates that blockchain technology extends beyond cryptocurrency applications to solve fundamental challenges in distributed systems coordination.

Acknowledgments

We thank the WalletConnect community and the 20+ independent node operators who participate in running WCN 2.0. Special thanks to the engineering teams who contributed to the design, implementation, and testing of this system.

References

[1] Elli Androulaki, Artem Barger, Vita Bortnikov, Christian Cachin, Konstantinos Christidis, Angelo De Caro, David Enyeart, Christopher Ferris, Gennady Laventman, Yacov Manevich, Srinivasan Muralidharan, Chet Murthy, Binh Nguyen,

Manish Sethi, Gari Singh, Keith Smith, Alessandro Sorniotti, Chrysoula Stathakopoulou, Marko Vukolić, Sharon Weed Cocco, and Jason Yellick. Hyperledger fabric: A distributed operating system for permissioned blockchains. In *Proceedings of the Thirteenth EuroSys Conference (EuroSys '18)*, pages 30:1–30:15. ACM, 2018.

[2] Cockroach Labs. Cockroachdb: The resilient geo-distributed sql database. https://www.cockroachlabs.com, 2020. Accessed: 2025-01-15.

[3] James C. Corbett, Jeffrey Dean, Michael Epstein, Andrew Fikes, Christopher Frost, J. J. Furman, Sanjay Ghemawat, Andrey Gubarev, Christopher Heiser, Peter Hochschild, Wilson Hsieh, Sebastian Kanthak, Eugene Kogan, Hongyi Li, Alexander Lloyd, Sergey Melnik, David Mwaura, David Nagle, Sean Quinlan, Rajesh Rao, Lindsay Rolig, Yasushi Saito, Michal Szymaniak, Christopher Taylor, Ruth Wang, and Dale Woodford. Spanner: Google's globally-distributed database. In *Proceedings of the 10th USENIX Symposium on Operating Systems Design and Implementation (OSDI '12)*, pages 251–264. USENIX Association, 2012.

[4] Giuseppe DeCandia, Deniz Hastorun, Madan Jampani, Gunavardhan Kakulapati, Avinash Lakshman, Alex Pilchin, Swaminathan Sivasubramanian, Peter Vosshall, and Werner Vogels. Dynamo: Amazon's highly available key-value store. *ACM SIGOPS Operating Systems Review*, 41(6):205–220, 2007.

[5] etcd Project. etcd: A distributed, reliable key-value store. https://etcd.io, 2013. Accessed: 2025-01-15.

[6] Avinash Lakshman and Prashant Malik. Cassandra: A decentralized structured storage system. *ACM SIGOPS Operating Systems Review*, 44(2):35–40, 2010.

[7] Leslie Lamport. The part-time parliament. *ACM Transactions on Computer Systems*, 16(2):133–169, 1998.

[8] Diego Ongaro and John Ousterhout. In search of an understandable consensus algorithm. In *Proceedings of the 2014 USENIX Annual Technical Conference (USENIX ATC '14)*, pages 305–319, Philadelphia, PA, 2014. USENIX Association.